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SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Packet Tracer - WLAN Configuration

Addressing Table

Device	Interface	IP Address
Home Wireless Router	Internet	DHCP
	LAN	192.168.6.1/27
RTR-1	G0/0/0.2	192.168.2.1/24
	G0/0/0.5	192.168.5.1/24
	G0/0/0.100	192.168.100.1/24
	G0/0/1	10.6.0.1/24
SW1	VLAN 200	192.168.100.100/24
LAP-1	G0	DHCP
WLC-1	Management	192.168.100.254/24
RADIUS Server	NIC	10.6.0.254/24
Home Admin	NIC	DHCP
Enterprise Admin	NIC	192.168.100.200/24
Web Server	NIC	203.0.113.78/24
DNS Server	NIC	10.100.100.252
Laptop	NIC	DHCP
Tablet PC	Wireless0	DHCP
Smartphone	Wireless0	DHCP
Wireless Host 1	Wireless0	DHCP

Device	Interface	IP Address
Wireless Host 2	Wireless0	DHCP

WLAN Information

WLAN	SSID	Authentication	Username	Password
Home Network	HomeSSID	WPA2-Personal	N/A	Cisco123
WLAN VLAN 2	SSID-2	WPA-2 Personal	N/A	Cisco123
WLAN VLAN 5	SSID-5	WPA-2 Enterprise	userWLAN5	userW5pass

Note: It is not a good practice to reuse passwords as is done in this activity. Passwords have been reused to make it easier to work through the tasks.

Objectives

In this activity, you will configure both a wireless home router and a WLC-based network. You will implement both WPA2-PSK and WPA2-Enterprise security.

Configure a home router to provide Wi-Fi connectivity to a variety of devices.

Configure WPA2-PSK security on a home router.

Configure interfaces on a WLC.

Configure WLANs on a WLC.

Configure WPA2-PSK security on a WLAN and connect hosts to WLAN.

Configure WPA2-Enterprise on a WLAN and connect hosts to the WLAN.

Verify connectivity WLAN connectivity.

Background / Scenario

You will apply your WLAN skills and knowledge by configuring a home wireless router and an enterprise WLC. You will implement both WPA2-PSK and WPA2-Enterprise security. Finally, you will connect hosts to each WLAN and verify connectivity.

Instructions

Part 1: Configure a Home Wireless Router.

You are installing a new home wireless router at a friend's house. You will need to change settings on the router to enhance security and meet your friend's requirements.

Step 1: Change DHCP settings.

- Open the Home Wireless Router GUI and change the router IP and DHCP settings according to the information in the Addressing Table.
- Permit a maximum of **20** addresses to be issued by the router.
- Configure the DHCP server to start with IP address **.3** of the LAN network.
- Configure the internet interface of the router to receive its IP address over DHCP.

Verify the address. What address did it receive?

- e. Configure the static DNS server to the address in the Addressing Table.

The screenshot shows the configuration interface for a 'Home Wireless Router'. The 'Config' tab is selected, and the 'Network Setup' section is active. The 'Internet Setup' section shows 'Automatic Configuration - DHCP' selected. The 'Network Setup' section shows the Router IP set to 192.168.6.1 and Subnet Mask set to 255.255.255.0. The DHCP Server is enabled, and the DHCP Reservation section is visible. The Static DNS section shows Static DNS 1 set to 10.100.100.252, and Static DNS 2, 3, and WINS are all set to 0.0.0.0.

Home Wireless Router

Physical Config **GUI** Attributes

Wireless Tri-Band Home Router

Firmware Ver

Setup Setup Wireless Security Access Restrictions Applications & Gaming Administration Status

Basic Setup DDNS MAC Address Clone Advanced Routing

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers):

Host Name: Domain Name: MTU: Size: 1500

Network Setup

Router IP: IP Address: 192 . 168 . 6 . 1 Subnet Mask: 255.255.255.0

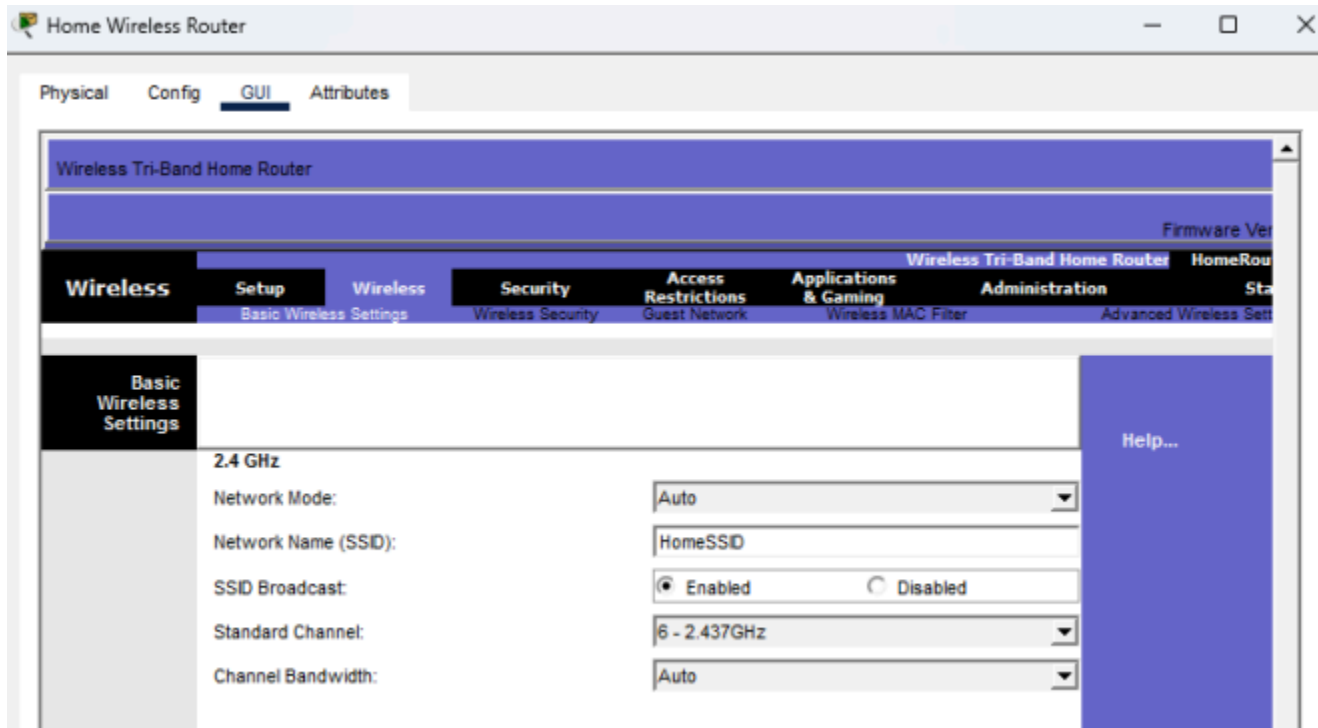
DHCP Server Settings: DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.0. 3 Maximum number of Users: 20 IP Address Range: 192.168.0. 3 - 22 Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 10 . 100 . 100 . 252 Static DNS 2: 0 . 0 . 0 . 0 Static DNS 3: 0 . 0 . 0 . 0 WINS: 0 . 0 . 0 . 0

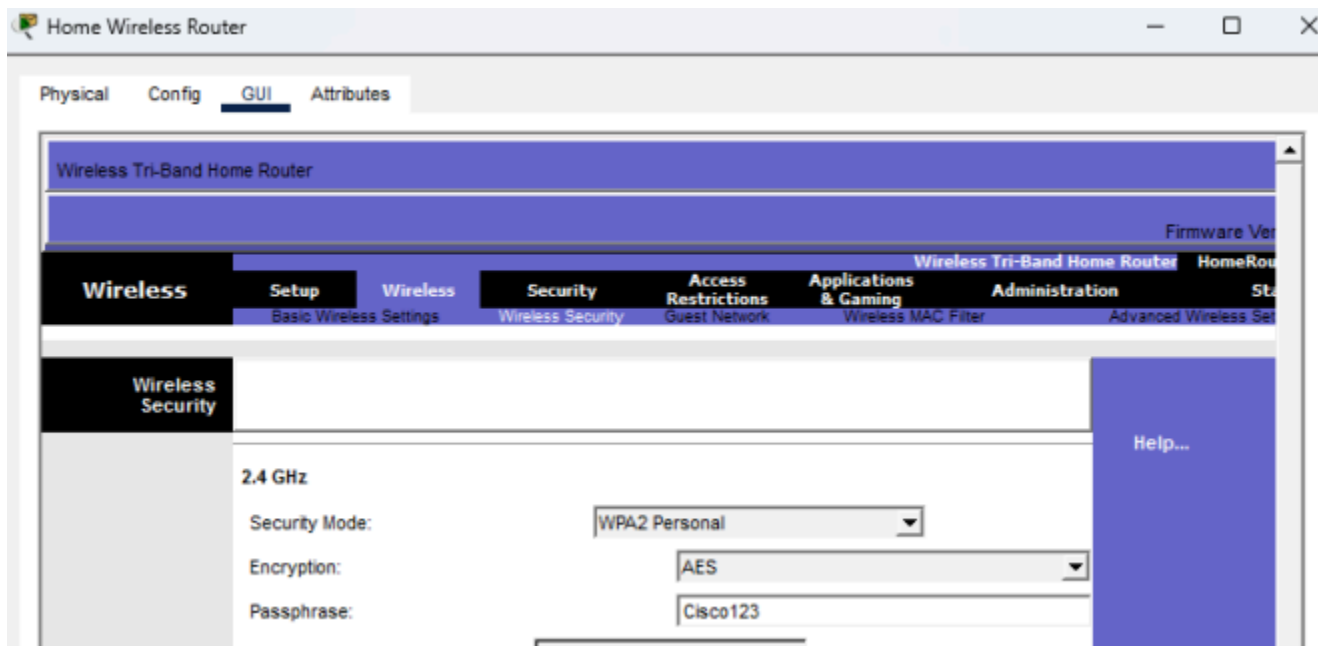
Step 2: Configure the Wireless LAN.

- The network will use the 2.4GHz Wireless LAN interface. Configure the interface with the SSID shown in the Wireless LAN information table.
- Use **channel 6**.
- Be sure that all wireless hosts in the home will be able to see the SSID.



Step 3: Configure security.

- Configure wireless LAN security. Use WPA2 Personal and the passphrase shown in the Wireless LAN information table.
- Secure the router by changing the default password to the value shown in the Wireless LAN information table.

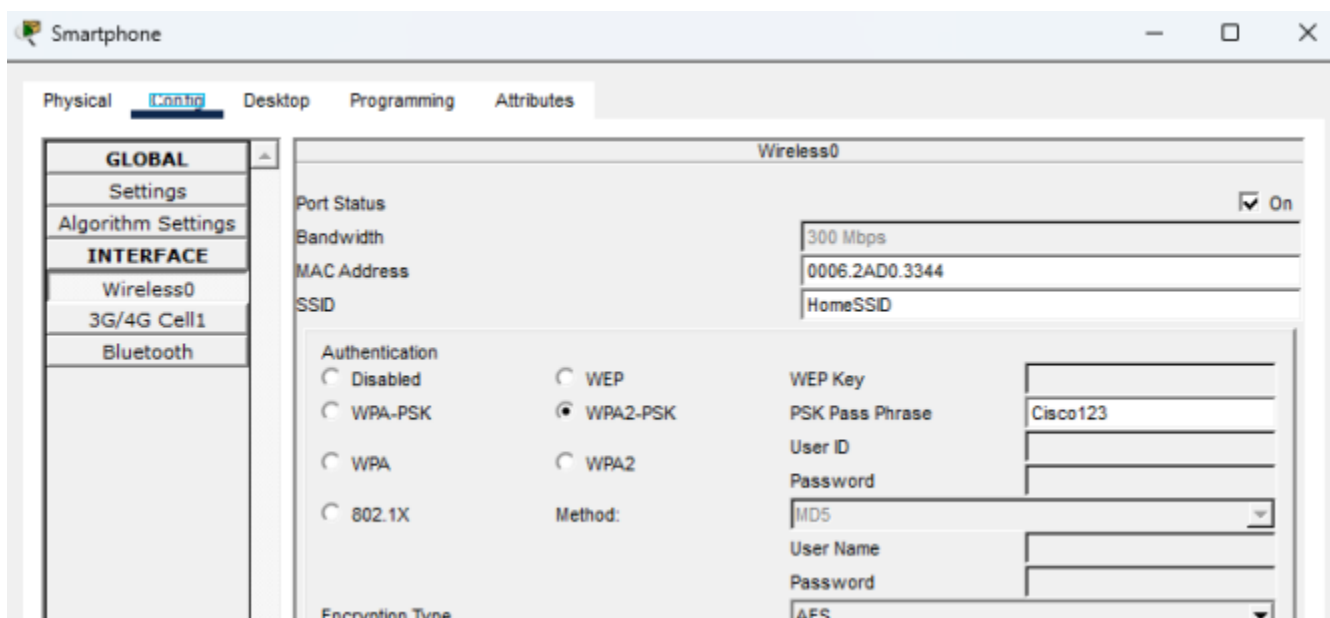
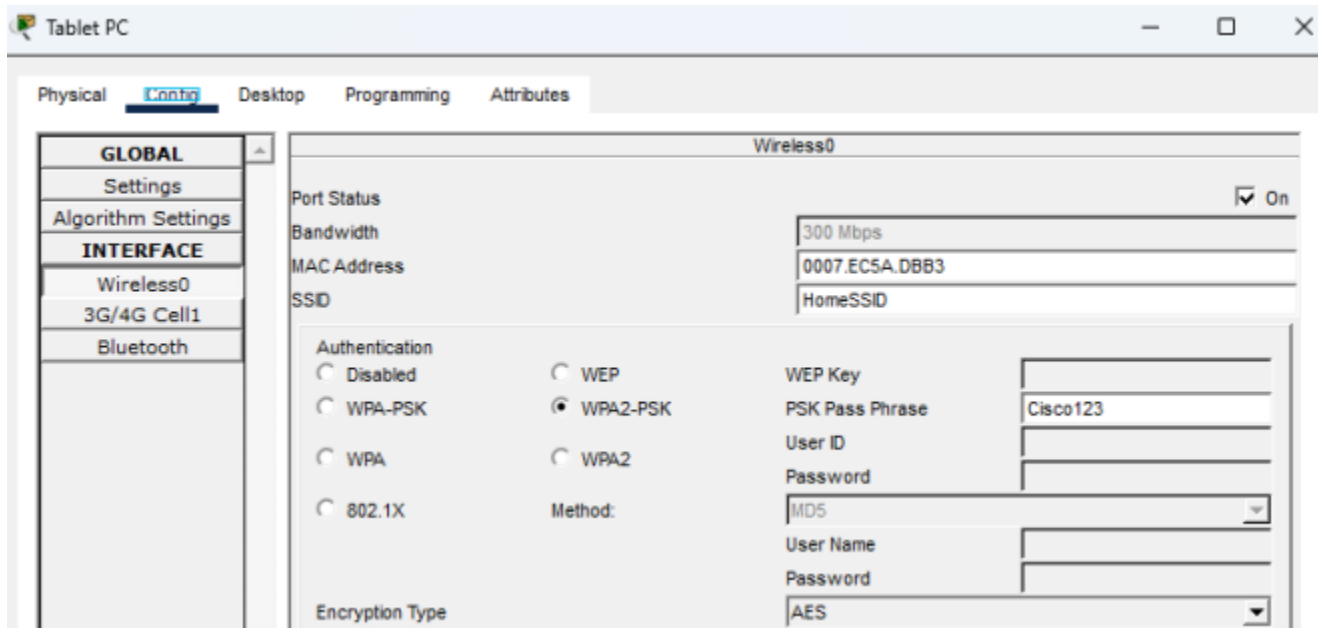


Step 4: Connect clients to the network.

- Open the PC Wireless app on the desktop of the laptop and configure the client to connect to the network.

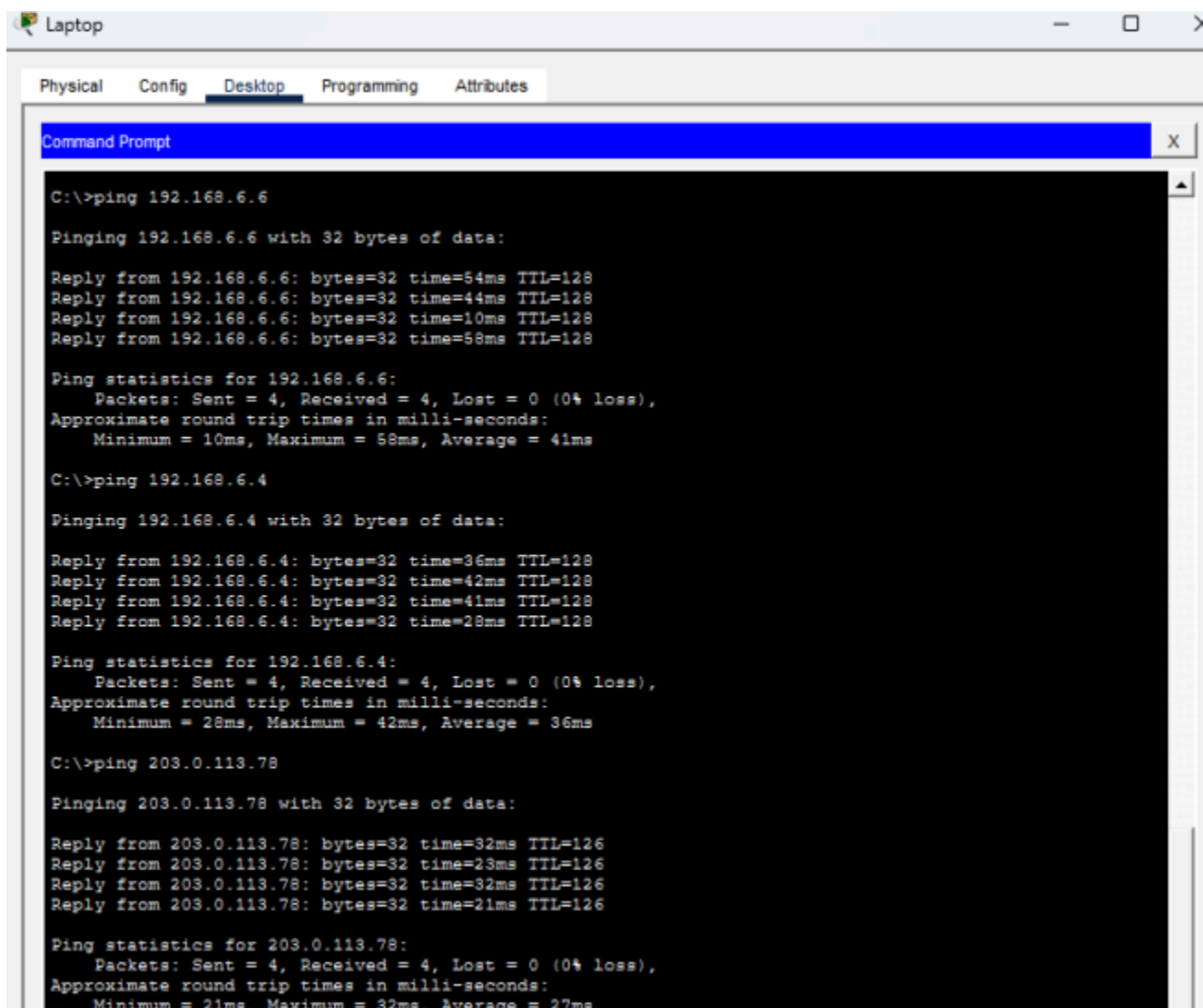
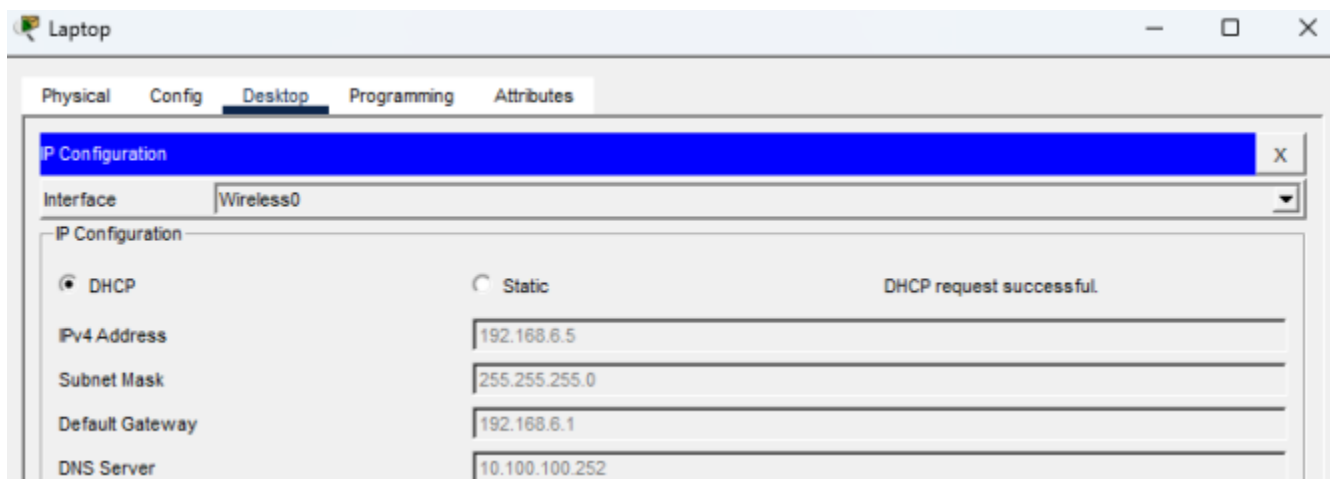


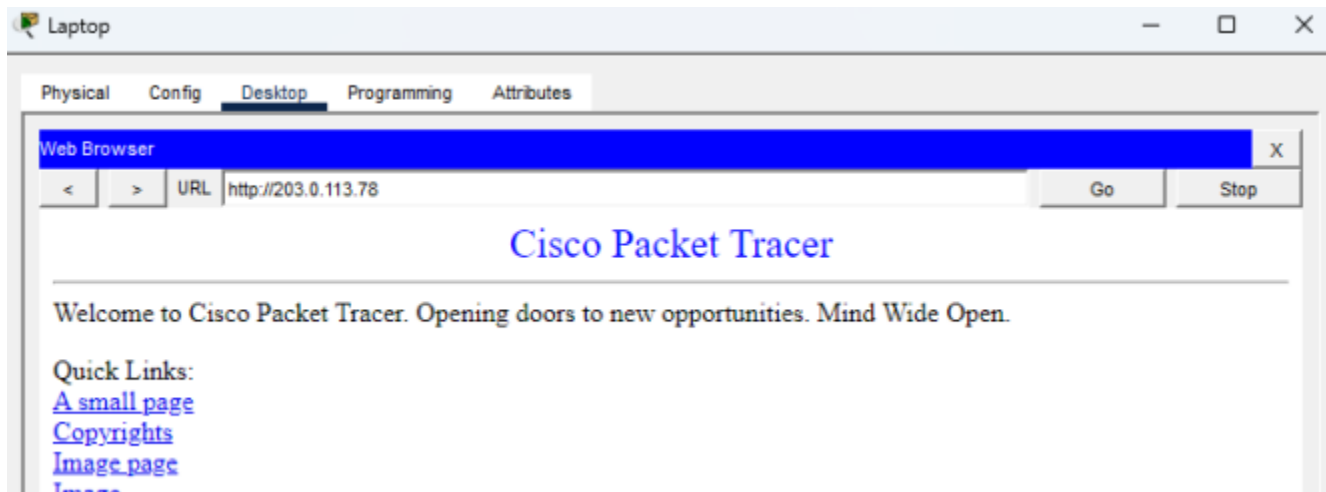
- b. Open the Config tab on the Tablet PC and Smartphone and configure the wireless interfaces to connect to the wireless network.



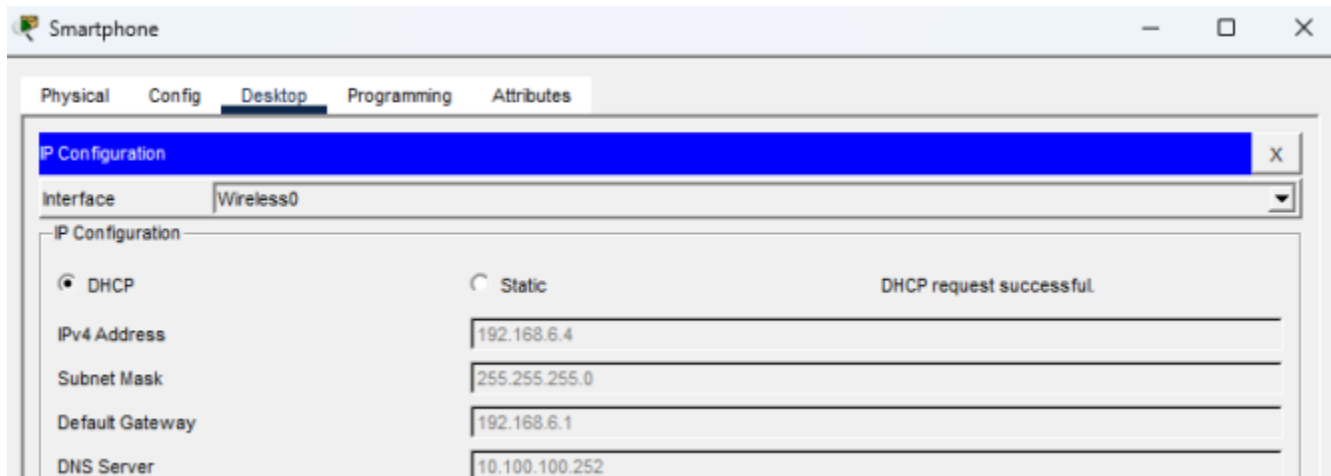
- c. Verify connectivity. The hosts should be able to ping each other and the web server. They should also be able to reach the web server URL.

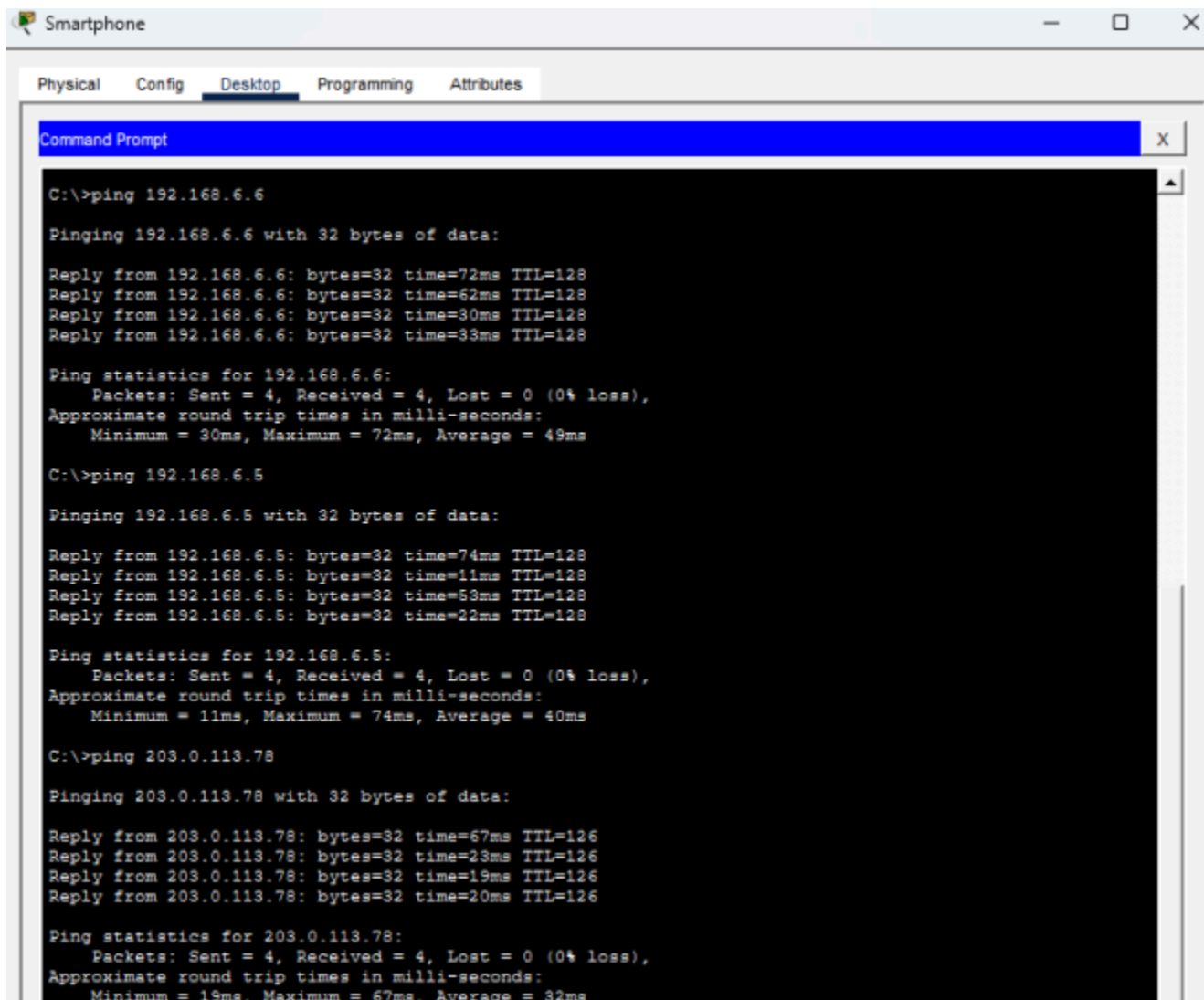
Laptop: Ping 192.168.6.6 (Tablet PC), 192.168.6.4 (Smartphone), 203.0.113.78 (Web Server)





Smartphone: Ping 192.168.6.6 (Tablet PC), 192.168.6.5 (Laptop), 203.0.113.78 (Web Server)





Smartphone

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.6.6

Pinging 192.168.6.6 with 32 bytes of data:

Reply from 192.168.6.6: bytes=32 time=72ms TTL=128
Reply from 192.168.6.6: bytes=32 time=62ms TTL=128
Reply from 192.168.6.6: bytes=32 time=30ms TTL=128
Reply from 192.168.6.6: bytes=32 time=33ms TTL=128

Ping statistics for 192.168.6.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 30ms, Maximum = 72ms, Average = 49ms

C:\>ping 192.168.6.5

Pinging 192.168.6.5 with 32 bytes of data:

Reply from 192.168.6.5: bytes=32 time=74ms TTL=128
Reply from 192.168.6.5: bytes=32 time=11ms TTL=128
Reply from 192.168.6.5: bytes=32 time=63ms TTL=128
Reply from 192.168.6.5: bytes=32 time=22ms TTL=128

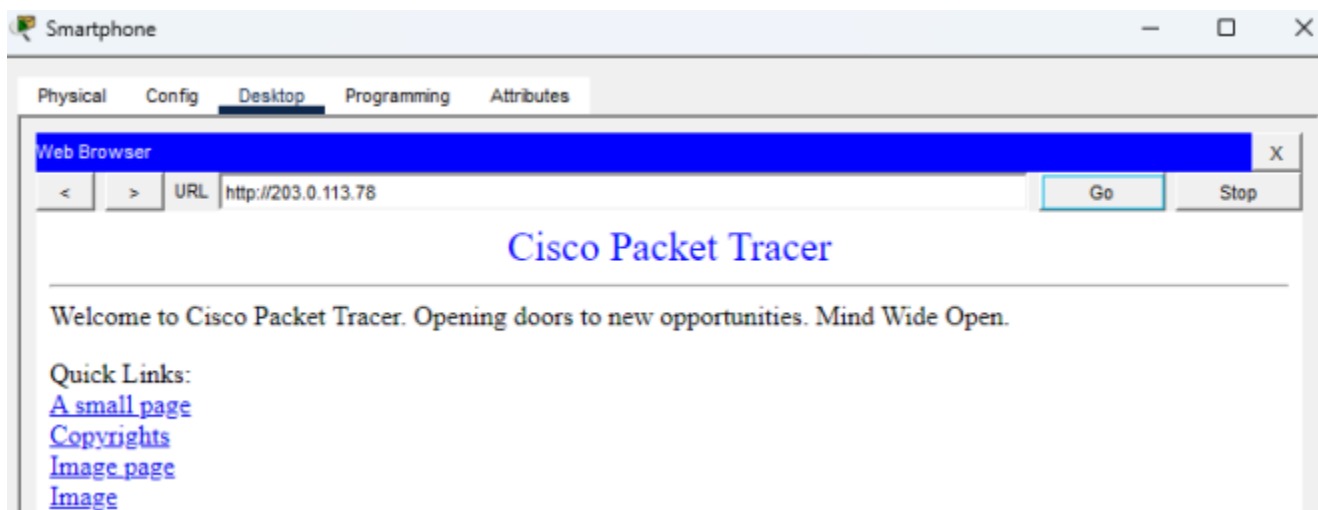
Ping statistics for 192.168.6.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 74ms, Average = 40ms

C:\>ping 203.0.113.78

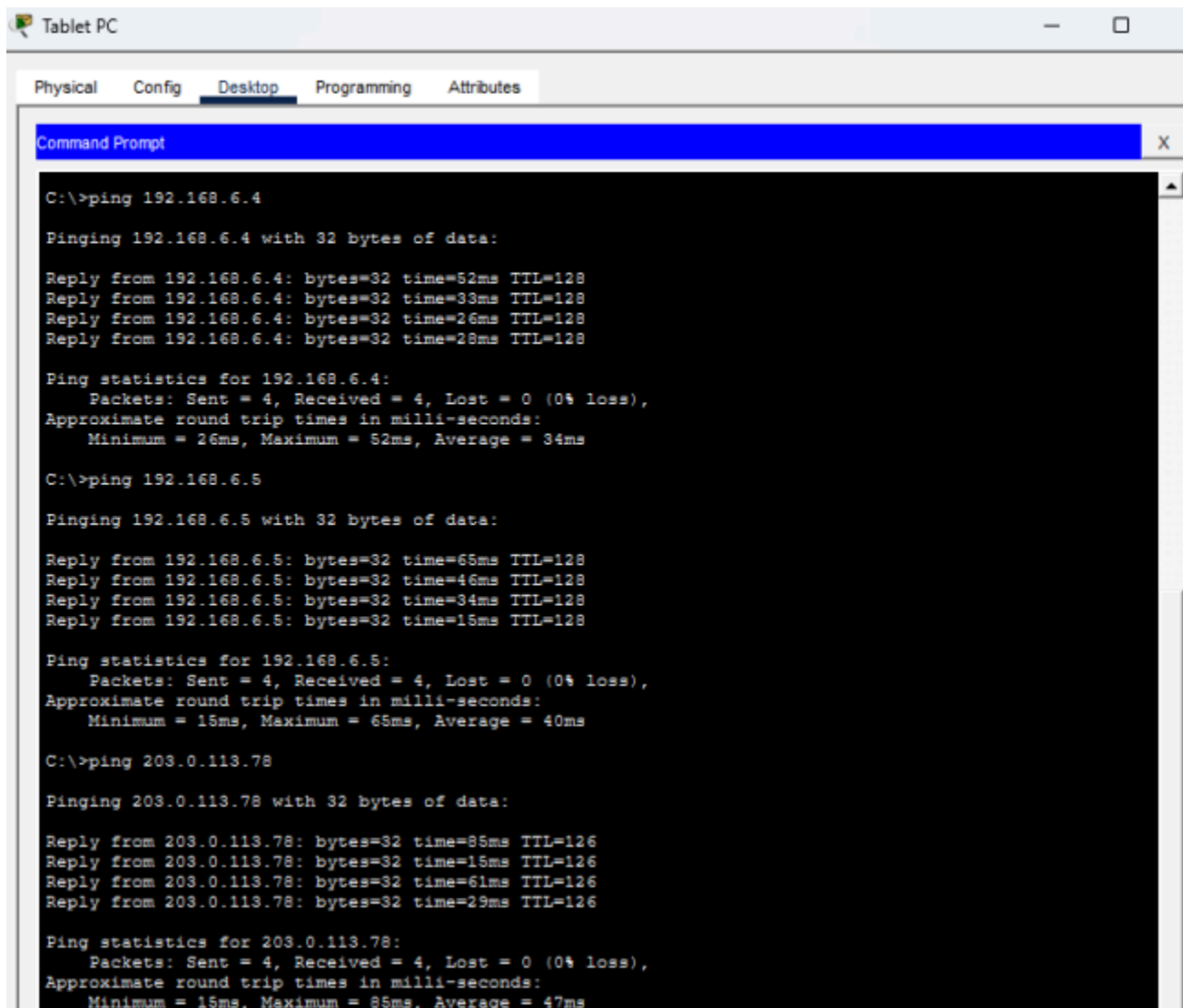
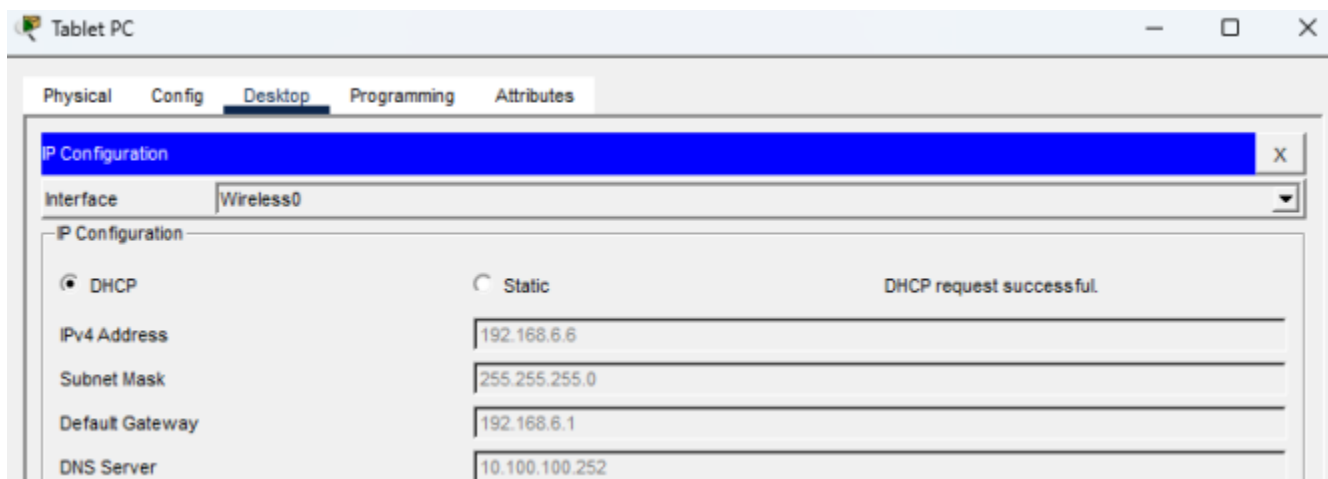
Pinging 203.0.113.78 with 32 bytes of data:

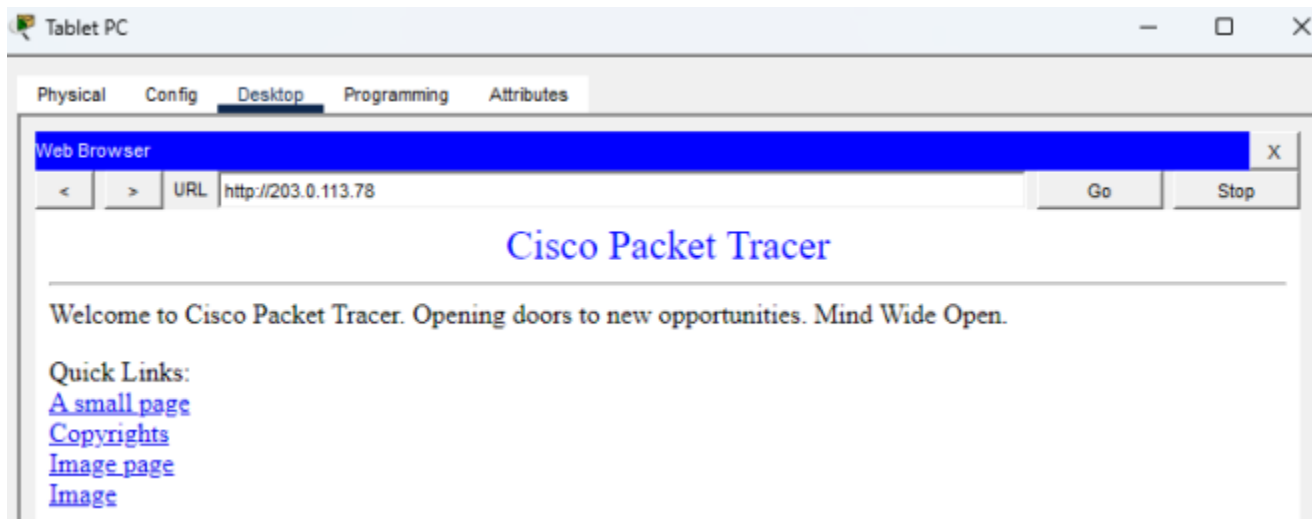
Reply from 203.0.113.78: bytes=32 time=67ms TTL=126
Reply from 203.0.113.78: bytes=32 time=23ms TTL=126
Reply from 203.0.113.78: bytes=32 time=19ms TTL=126
Reply from 203.0.113.78: bytes=32 time=20ms TTL=126

Ping statistics for 203.0.113.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 67ms, Average = 32ms
```



Tablet PC: Ping 192.168.6.4 (Smartphone), 192.168.6.5 (Laptop), 203.0.113.78 (Web Server)



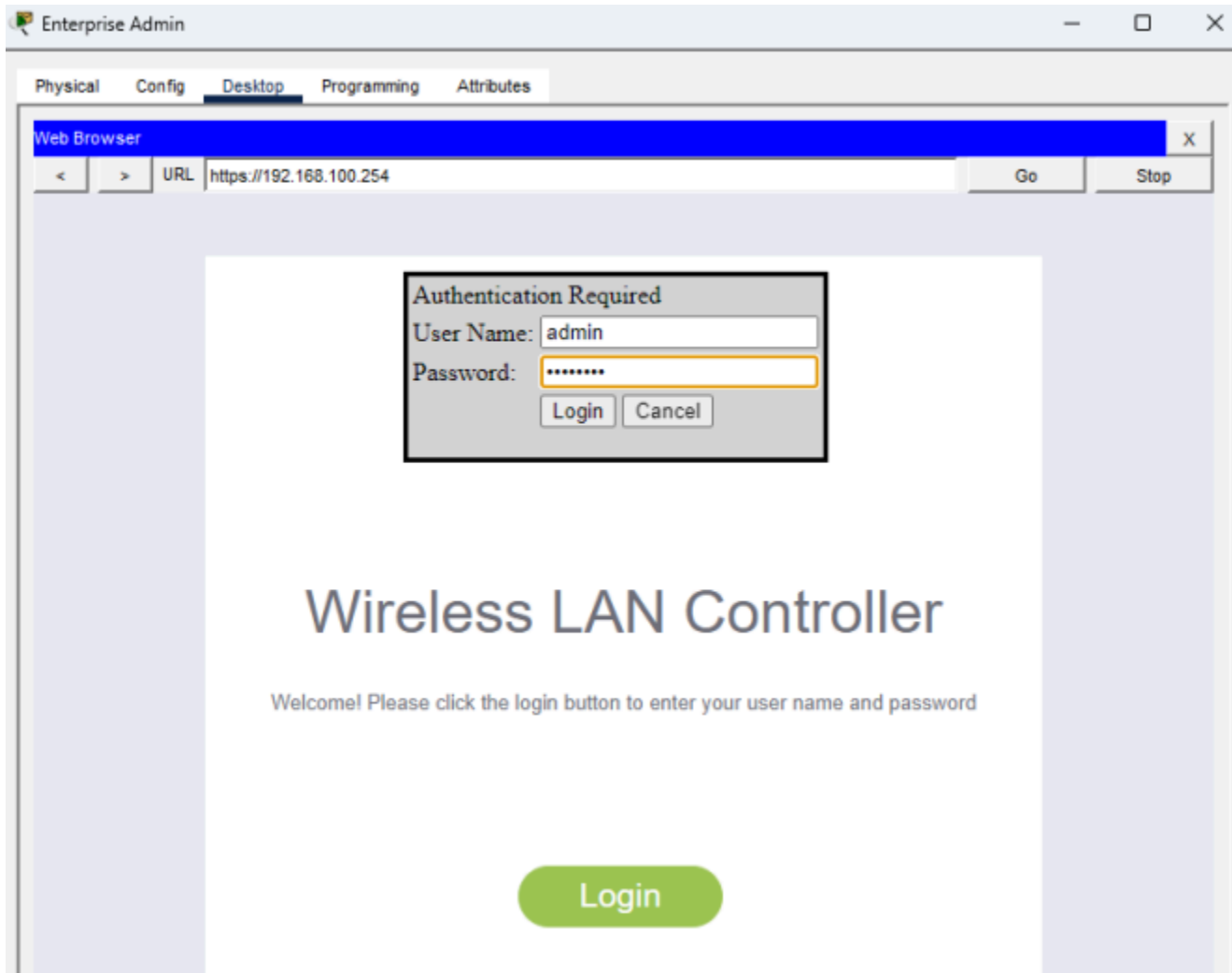


Part 2: Configure a WLC Controller Network

Configure the wireless LAN controller with two WLANs. One WLAN will use WPA2-PSK authentication. The other WLAN will use WPA2-Enterprise authentication. You will also configure the WLC to use an SNMP server and configure a DHCP scope that will be used by the wireless management network.

Step 1: Configure VLAN interfaces.

- a. From the Enterprise Admin, navigate to the WLC-1 management interface via a web browser. To log into WLC-1, use **admin** as the username and **Cisco123** as the password.



- b. Configure an interface for the first WLAN.
- Name: **WLAN 2**
 - VLAN Identifier: **2**
 - Port Number: **1**
 - Interface IP Address: **192.168.2.254**
 - Netmask: **255.255.255.0**
 - Gateway: **RTR-1 G0/0/0.2 address**
 - Primary DHCP Server: **Gateway address**

The screenshot shows the Cisco Enterprise Admin interface for configuring a WLAN. The left sidebar contains a navigation menu with the following items: General, Inventory, Interfaces, Interface Groups, Multicast, Internal DHCP Server, Mobility Management, Ports, NTP, CDP, Tunneling, IPv6, mDNS, and Advanced. The main content area is titled 'Controller' and displays the configuration for 'WLAN 2'. The configuration is organized into several sections: General Information (Interface Name: WLAN 2, MAC Address: 00:01:C7:C4:9D:BC), Configuration (Guest Lan, Quarantine, Quarantine Vlan Id: 0, NAS-ID), Physical Information (Port Number: 1, Backup Port: 0, Active Port: 1, Enable Dynamic AP Management), Interface Address (VLAN Identifier: 2, IP Address: 192.168.2.254, Netmask: 255.255.255.0, Gateway: 192.168.2.1), and DHCP Information (Primary DHCP Server: 192.168.2.1, Secondary DHCP Server).

General Information	
Interface Name	WLAN 2
MAC Address	00:01:C7:C4:9D:BC

Configuration	
Guest Lan	☐
Quarantine	☐
Quarantine Vlan Id	0
NAS-ID	

Physical Information	
Port Number	1
Backup Port	0
Active Port	1
Enable Dynamic AP Management	☐

Interface Address	
VLAN Identifier	2
IP Address	192.168.2.254
Netmask	255.255.255.0
Gateway	192.168.2.1

DHCP Information	
Primary DHCP Server	192.168.2.1
Secondary DHCP Server	

- c. Configure an interface for the second WLAN.

Name: **WLAN 5**

VLAN Identifier: **5**

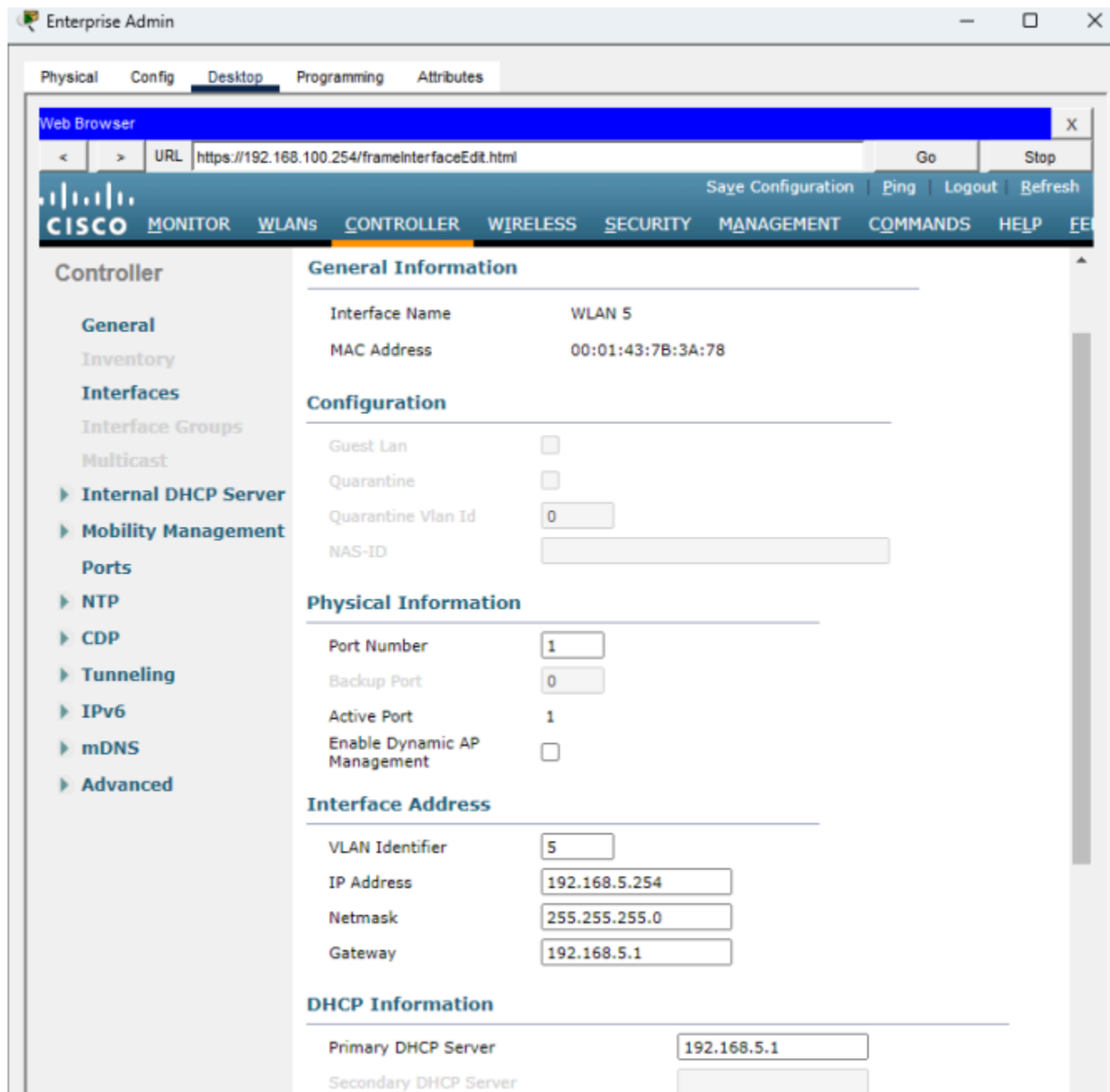
Port Number: **1**

Interface IP Address: **192.168.5.254**

Netmask: **255.255.255.0**

Gateway: **RTR-1 interface G0/0/0.5 address**

Primary DHCP Server: **Gateway address**



Enterprise Admin

Physical Config **Desktop** Programming Attributes

Web Browser
URL: <https://192.168.100.254/frameInterfaceEdit.html> Go Stop

Save Configuration | Ping | Logout | Refresh

CISCO MONITOR **WLANs** CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP

Controller

- General
- Inventory
- Interfaces
- Interface Groups
- Multicast
- Internal DHCP Server
- Mobility Management
- Ports
- NTP
- CDP
- Tunneling
- IPv6
- mDNS
- Advanced

General Information

Interface Name: WLAN 5
MAC Address: 00:01:43:7B:3A:78

Configuration

Guest Lan: ☐
Quarantine: ☐
Quarantine Vlan Id: 0
NAS-ID:

Physical Information

Port Number: 1
Backup Port: 0
Active Port: 1
Enable Dynamic AP Management: ☐

Interface Address

VLAN Identifier: 5
IP Address: 192.168.5.254
Netmask: 255.255.255.0
Gateway: 192.168.5.1

DHCP Information

Primary DHCP Server: 192.168.5.1
Secondary DHCP Server:

Step 2: Configure a DHCP scope for the wireless management network.

Configure and enable an internal DHCP scope as follows:

Scope Name: **management**

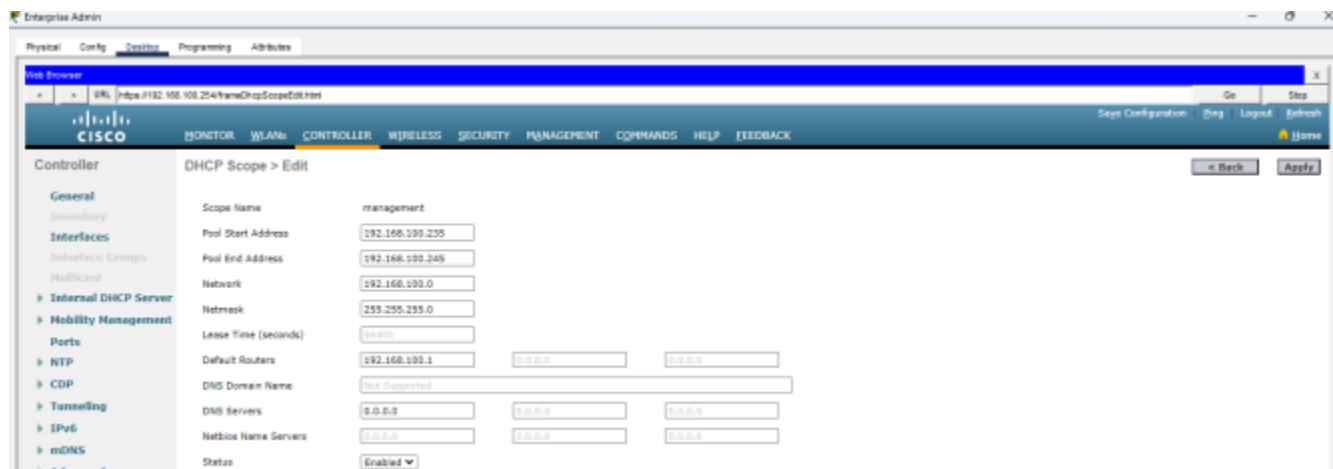
Pool Start Address: **192.168.100.235**

Pool End Address: **192.168.100.245**

Network: **192.168.100.0**

Netmask: **255.255.255.0**

Default Routers: **192.168.100.1**



Step 3: Configure the WLC with external server addresses.

- a. Configure the RADIUS server information as follows:

Sever Index: 1

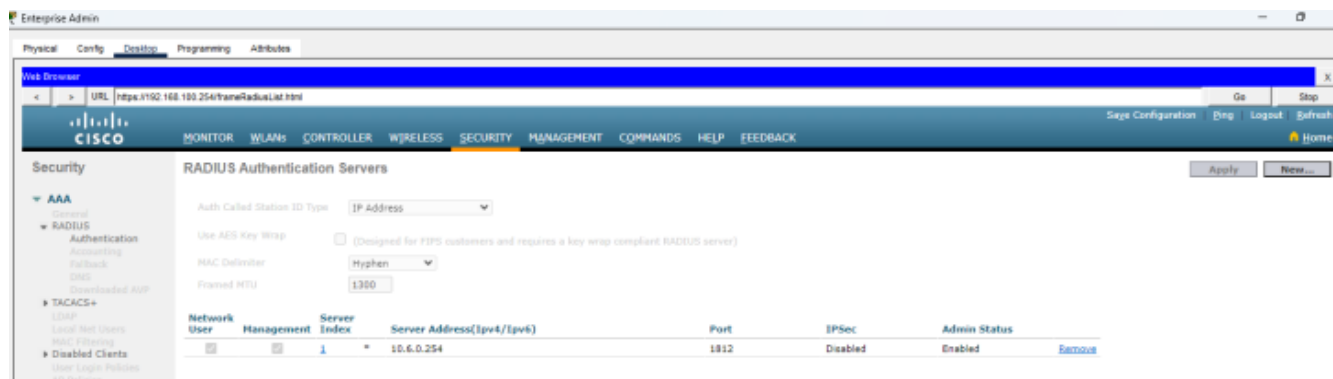
Sever Address: **10.6.0.254**

Shared Secret: **RadiusPW**

- b. Configure the WLC to send logs information to an SNMP server.

Community Name: **WLAN**

IP Address: **10.6.0.254**



Step 4: Create the WLANs.

- a. Create the first WLAN:

Profile Name: **Wireless VLAN 2**

WLAN SSID: **SSID-2**

ID: **2**

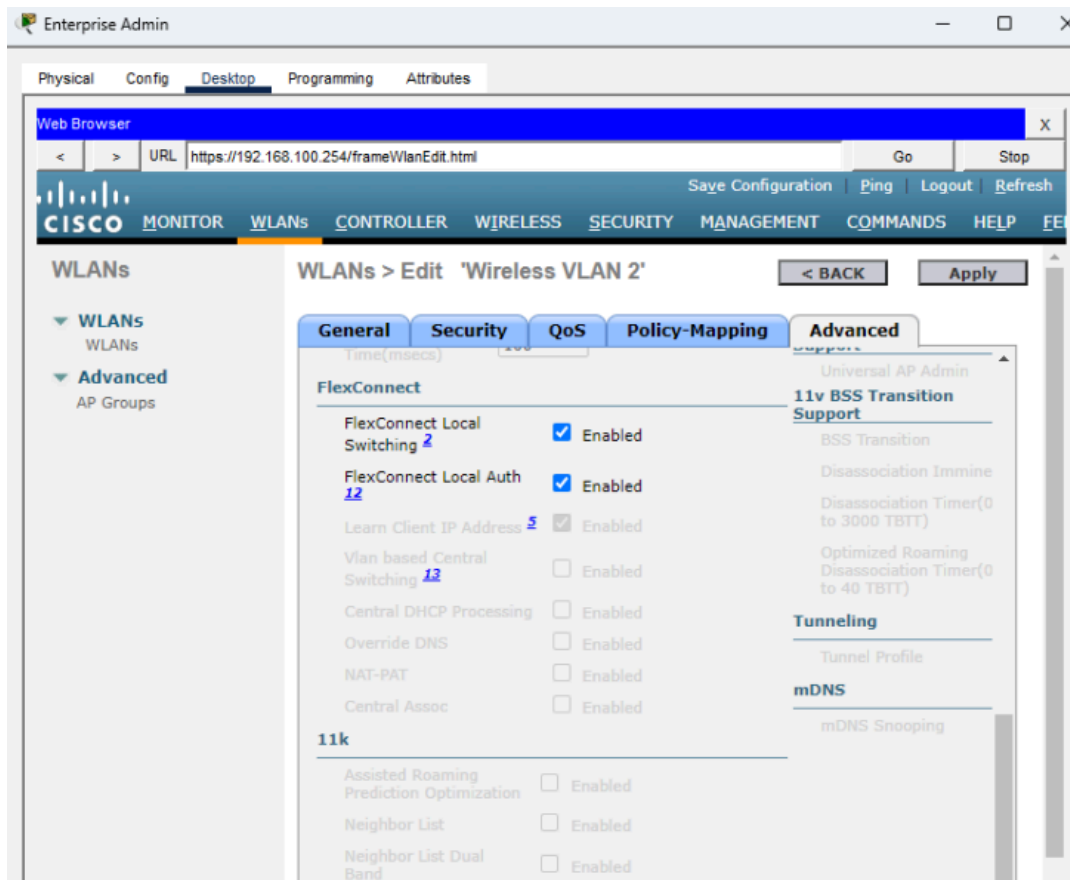
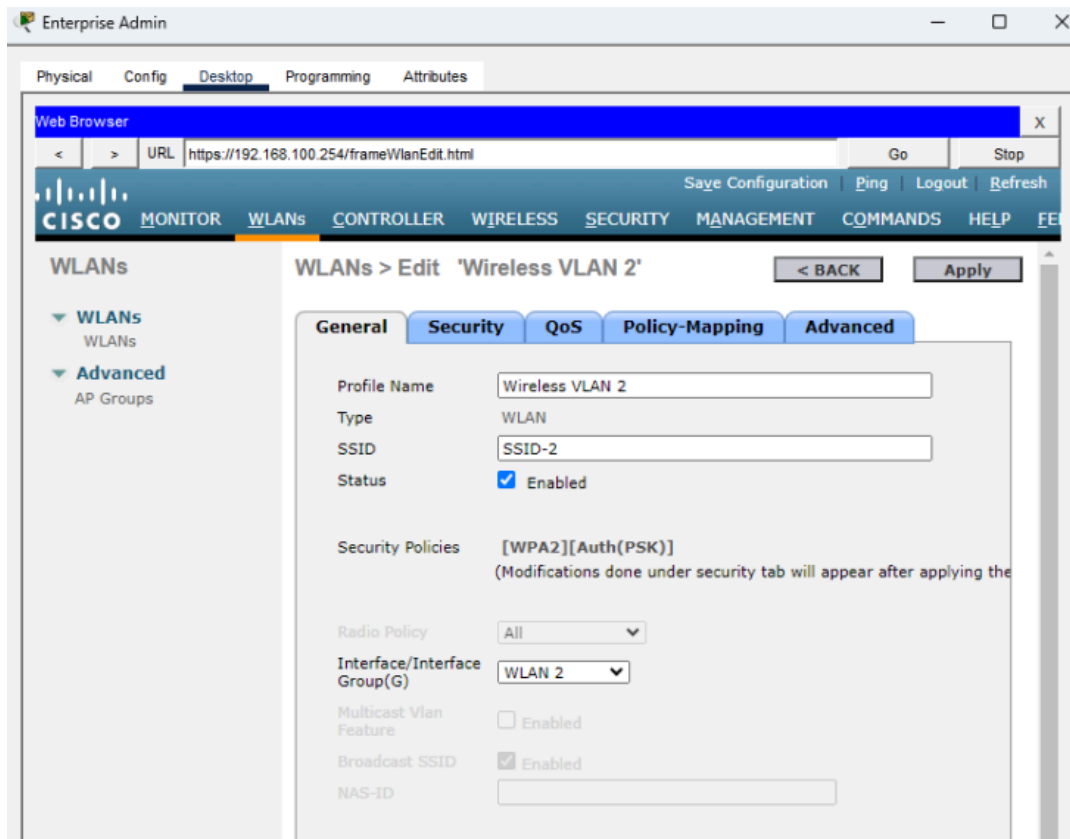
Interface: **WLAN 2**

Security: **WPA2-PSK**

Passphrase: **Cisco123**

Under the Advanced tab, go to the FlexConnect section. Enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.

Packet Tracer - WLAN Configuration



- b. Create the second WLAN:

Profile Name: **Wireless VLAN 5**

WLAN SSID: **SSID-5**

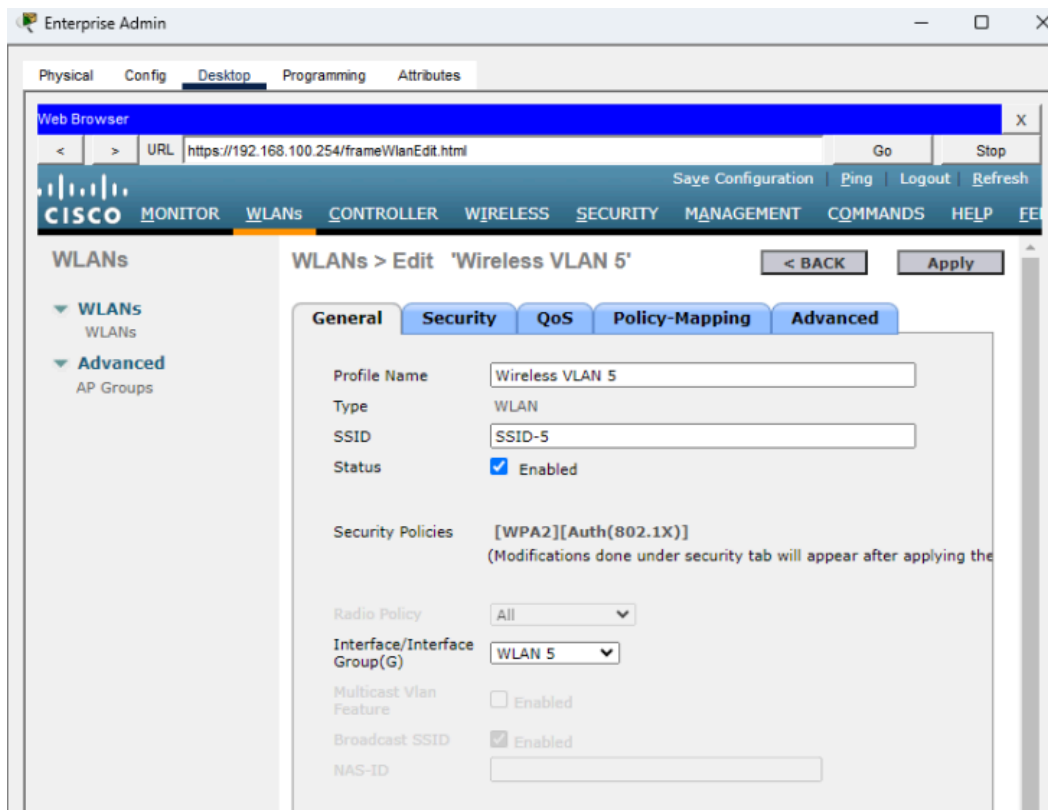
Interface: **WLAN 5**

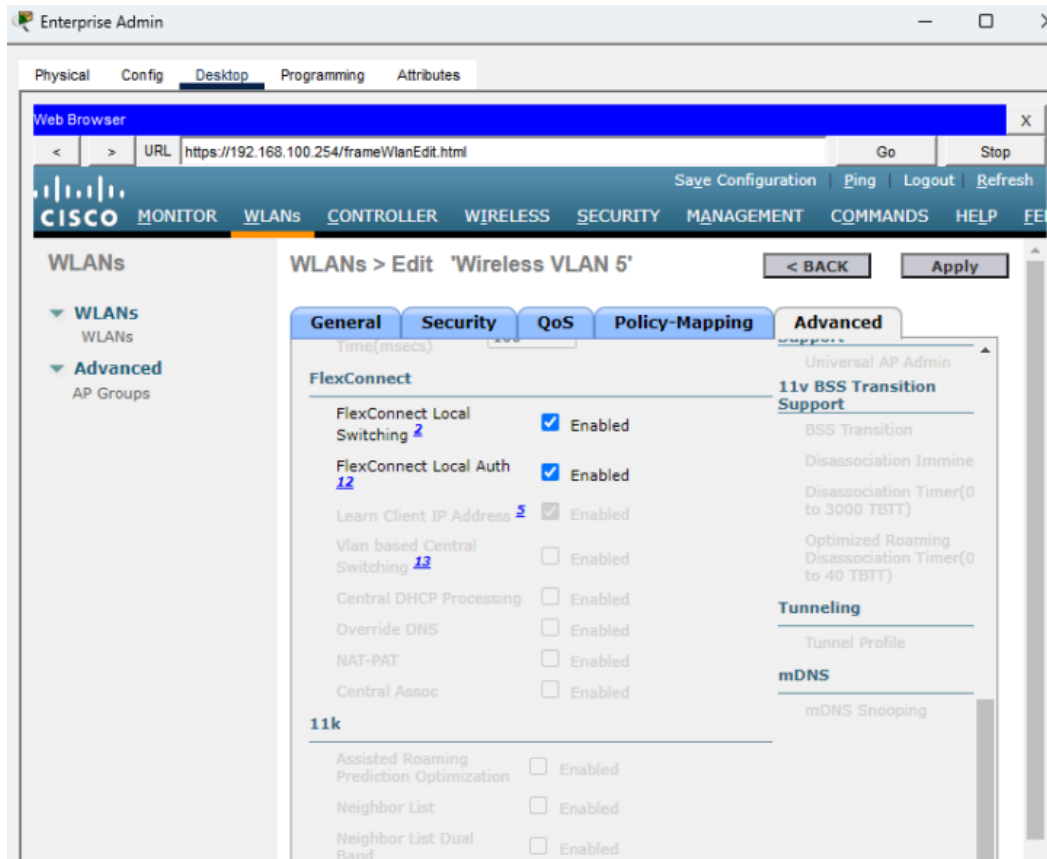
ID: **5**

Security: **802.1x - WPA2-Enterprise**

Configure the WLAN to use the RADIUS server for authentication.

Make the **FlexConnect** settings as was done in Step 4a.





Step 5: Configure the hosts to connect to the WLANs.

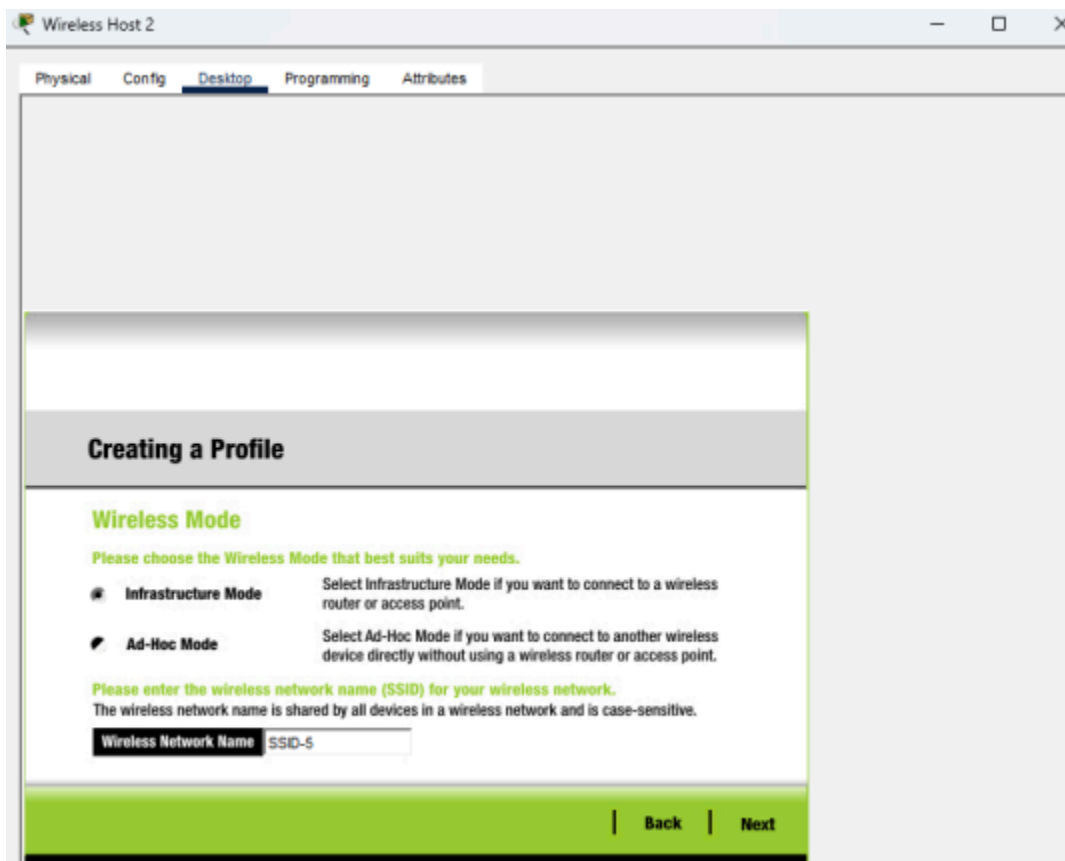
Use the desktop PC Wireless app to configure the hosts as follows:

- Wireless Host 1 should connect to Wireless VLAN 2.



b. Wireless Host 2 should connect to Wireless VLAN 5 using the credentials in the WLAN information table.





Wireless Host 2

Physical Config **Desktop** Programming Attributes

Creating a Profile

Wireless Mode

Please choose the Wireless Mode that best suits your needs.

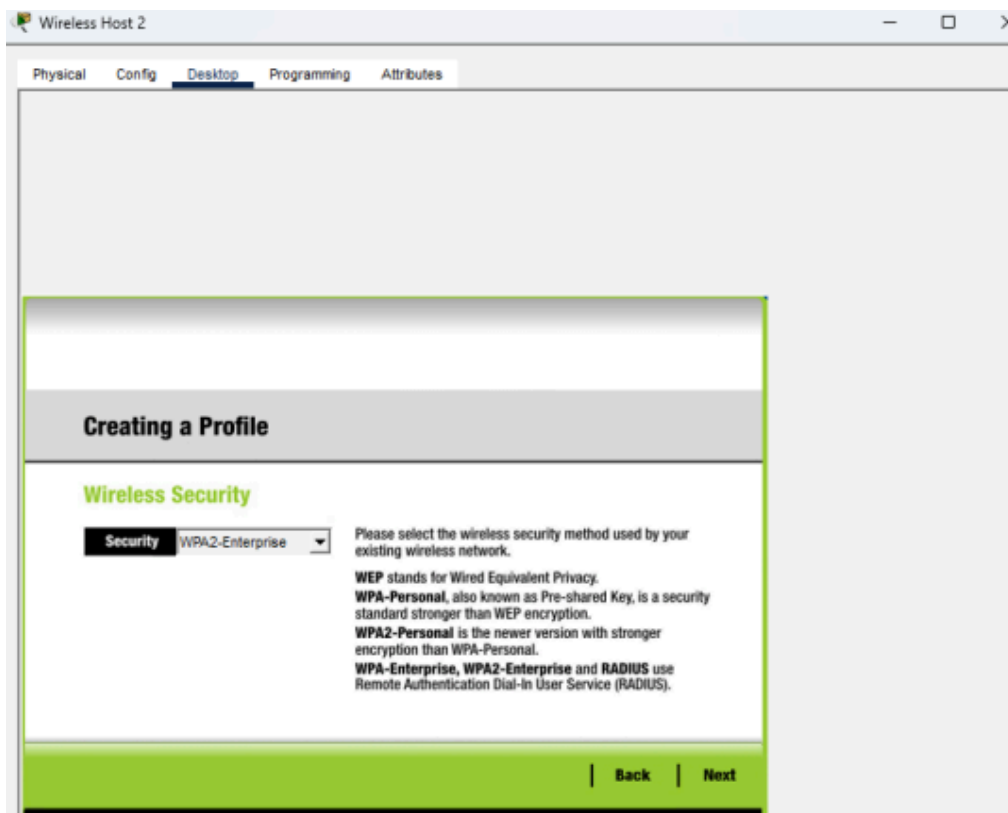
☒ **Infrastructure Mode** Select Infrastructure Mode if you want to connect to a wireless router or access point.

☐ **Ad-Hoc Mode** Select Ad-Hoc Mode if you want to connect to another wireless device directly without using a wireless router or access point.

Please enter the wireless network name (SSID) for your wireless network.
The wireless network name is shared by all devices in a wireless network and is case-sensitive.

Wireless Network Name

Back Next



Wireless Host 2

Physical Config **Desktop** Programming Attributes

Creating a Profile

Wireless Security

Security

Please select the wireless security method used by your existing wireless network.

WEP stands for Wired Equivalent Privacy.
WPA-Personal, also known as Pre-shared Key, is a security standard stronger than WEP encryption.
WPA2-Personal is the newer version with stronger encryption than WPA-Personal.
WPA-Enterprise, WPA2-Enterprise and RADIUS use Remote Authentication Dial-In User Service (RADIUS).

Back Next

Wireless Host 2

Physical Config Desktop Programming Attributes

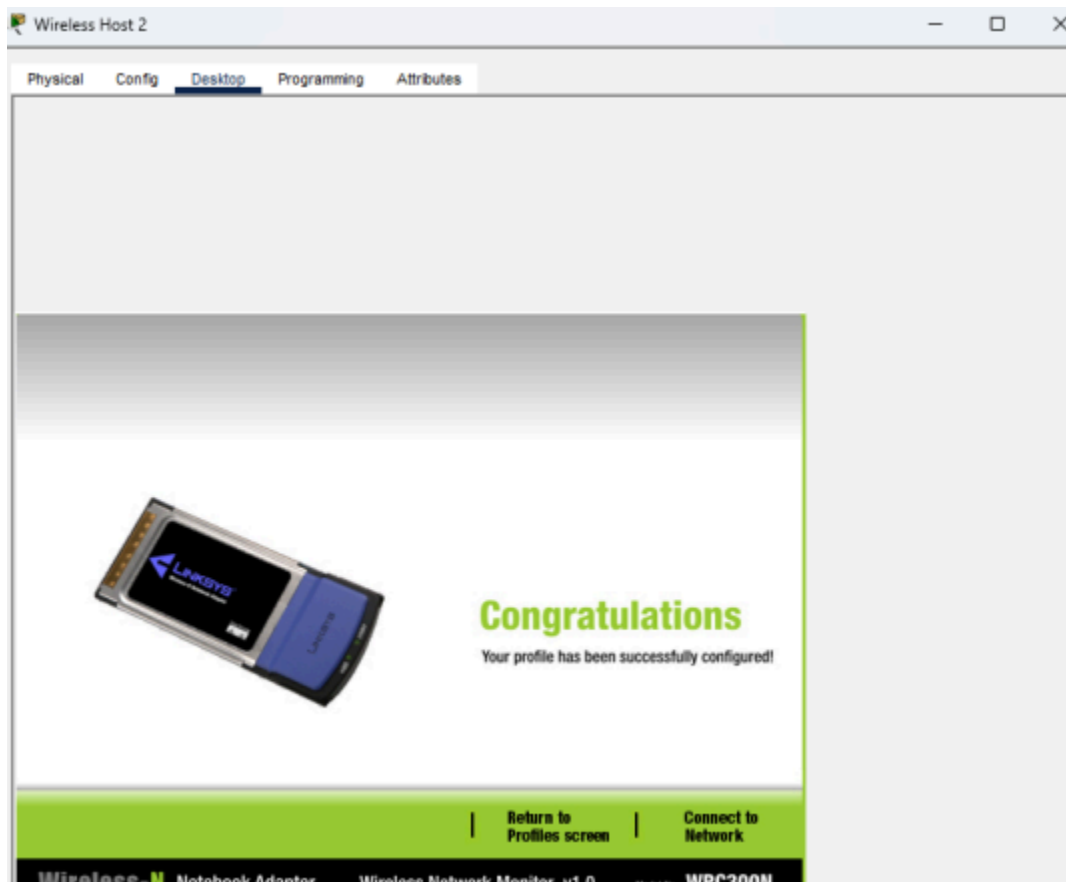
Creating a Profile

Wireless Security - WPA2 Enterprise

Authentication	PEAP	Please select the authentication method that you use to access your network.
Login Name	userWLAN5	Enter the Login Name used for authentication.
Password	*****	Enter the Password used for authentication.
Server Name		Enter the Server Name used for authentication. (Optional)
Certificate	Trust Any	Please select the certificate used for authentication.
Inner Authen.	TOKEN CARD	Please select the inner authentication method used inside the PEAP tunnel.

Back Next

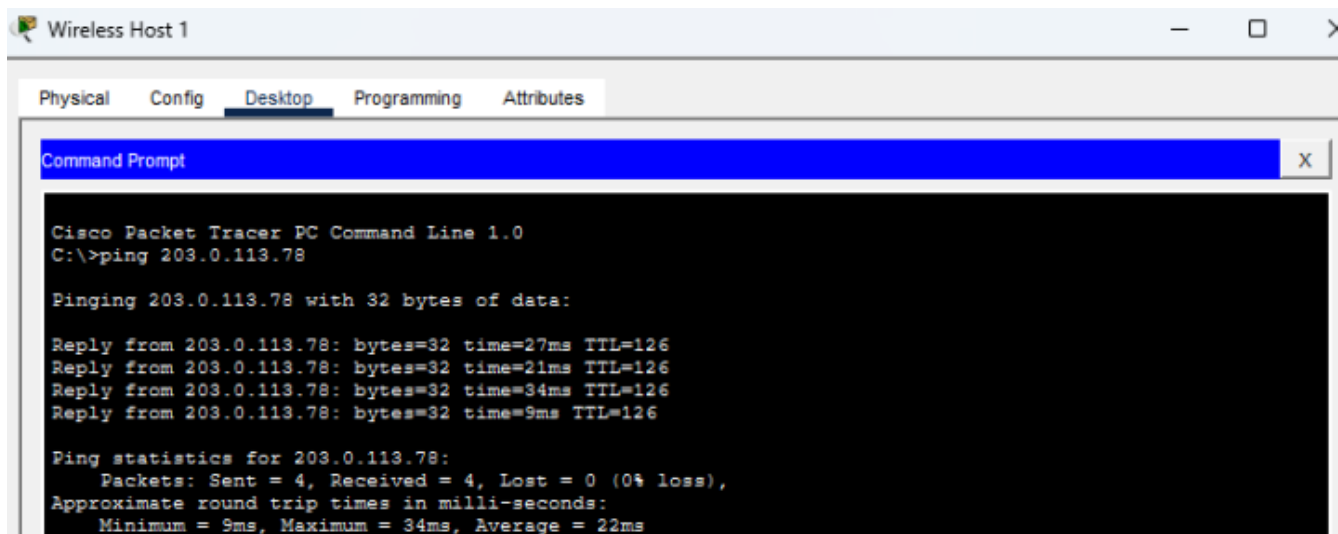
Wireless-N Notebook Adapter Wireless Network Monitor v111 WPC300N

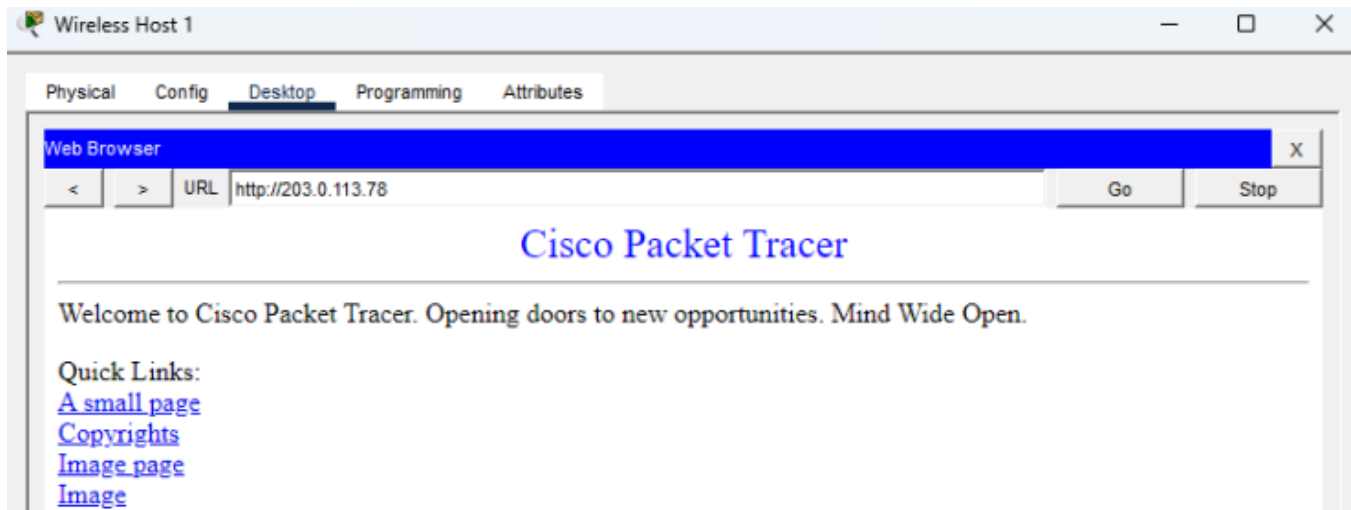


Step 6: Test connectivity.

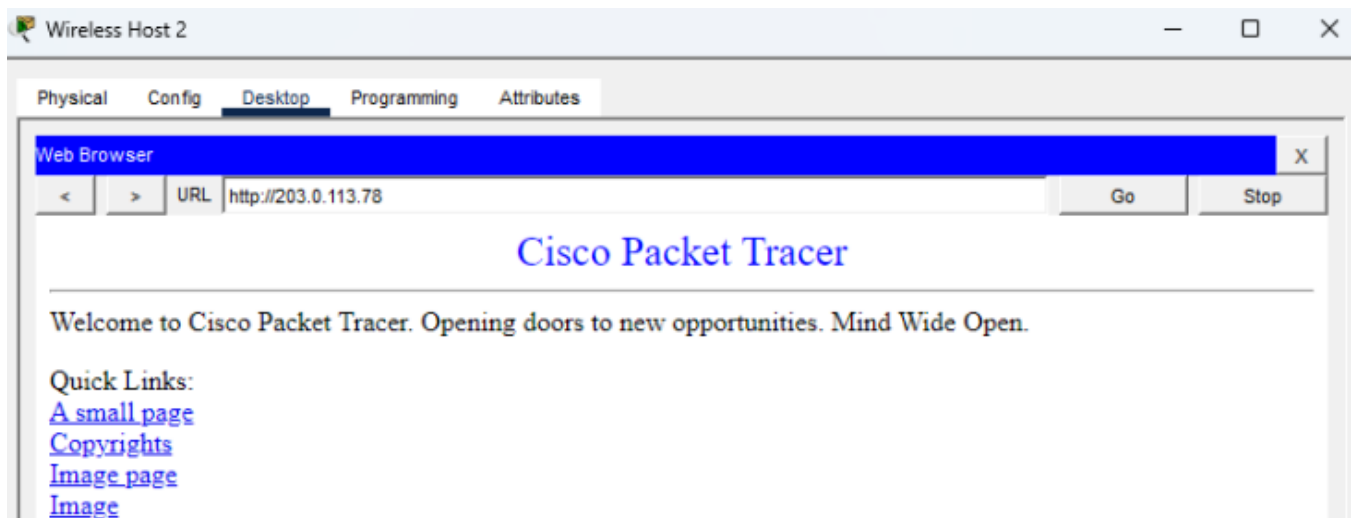
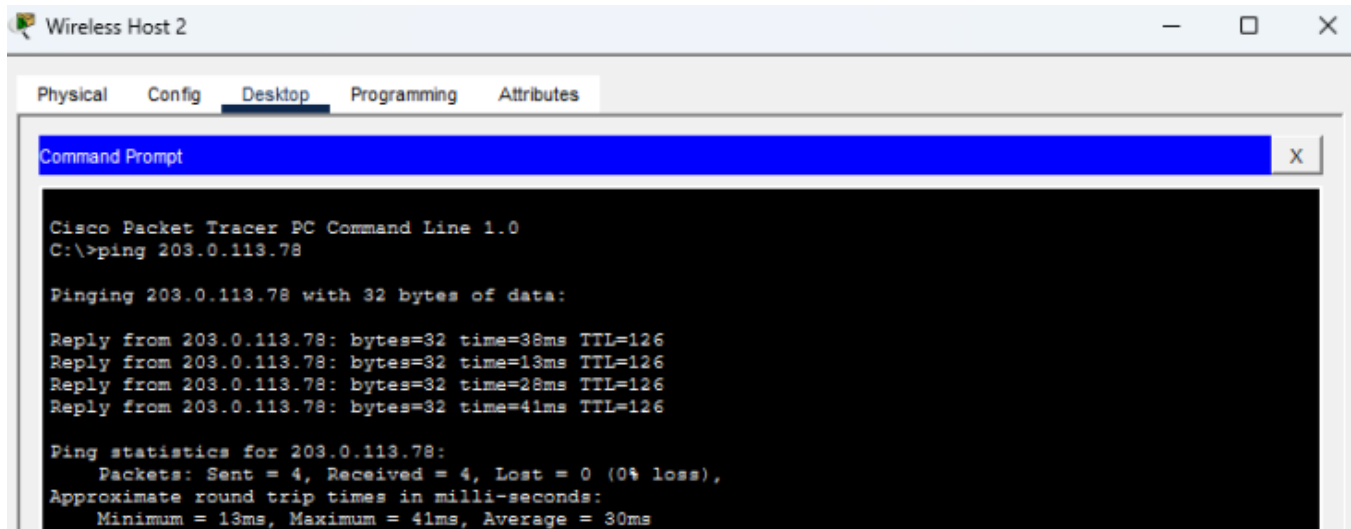
Test connectivity between the wireless hosts and the Web Server by ping and URL.

Wireless Host 1: Ping 203.0.113.78 (Web Server)





Wireless Host 2: Ping 203.0.113.78 (Web Server)



Topology

